

Description

The project was developed using a set of powerful Python-based tools and libraries for data processing, machine learning, visualization, and deployment. These technologies provide the foundation for efficient model development, analysis, and application deployment.

Anaconda Navigator

A desktop graphical user interface used to launch applications and manage Python packages and environments for data science.

Official Website: <https://www.anaconda.com/download>

PyCharm

An Integrated Development Environment (IDE) specifically designed for Python programming, offering intelligent code completion, debugging, and project management features.

Official Website: <https://www.jetbrains.com/pycharm/>

NumPy

A fundamental Python library for numerical computing that provides support for large, multi-dimensional arrays, matrices, and high-performance mathematical operations.

Official Documentation: <https://numpy.org/doc/stable/>

Pandas

An open-source Python library used for data manipulation and analysis, offering powerful data structures such as DataFrames for handling structured data.

Official Documentation: <https://pandas.pydata.org/docs/>

Scikit-learn

A popular machine learning library in Python that provides algorithms for classification, regression, clustering, model evaluation, and data preprocessing.

Official Documentation: <https://scikit-learn.org/stable/>

Matplotlib

A comprehensive plotting library used to create static, animated, and interactive visualizations in Python for data analysis and reporting.

Seaborn

Official Documentation: <https://matplotlib.org/stable/>

A statistical data visualization library built on top of Matplotlib that enables the creation of attractive and informative graphical representations.

Official Documentation: <https://seaborn.pydata.org/>

Flask

A lightweight Python web framework used to build web applications and deploy machine learning models through a simple and flexible interface.

Official Documentation: <https://flask.palletsprojects.com/>